

Parvez Akhtar

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PROFESSIONAL SUMMARY

Senior AI/ML Engineer with 5+ years deploying machine learning into production and building production-grade RAG (Retrieval-Augmented Generation) pipelines, agentic workflows, and LLM applications. Sets technical direction end to end: fine-tuning and evaluating models, enforcing guardrails and prompt-injection defense for safe, deterministic outputs, and running low-latency inference on scalable MLOps pipelines. Reduced inference costs by ~90% while taking experimental AI to reliable enterprise production in small, fast-moving teams.

TECHNICAL SKILLS

GenAI & LLM Applications: RAG (Retrieval-Augmented Generation) Pipelines, Agentic Workflows & Multi-Agent Orchestration (LangChain, LlamaIndex, LangGraph, OpenAI Agents SDK, Anthropic SDK), Tool/Function Calling, Structured Outputs, Fine-tuning (LoRA/QLoRA, Hugging Face), Prompt Engineering, MCP (Model Context Protocol).

AI Trust & Alignment: Guardrails & Prompt-Injection Defense, Evaluation Frameworks (RAGAS, Langfuse), CI/CD Evaluation Gates, Safety Alignment, Deterministic & Validated Outputs.

ML/DL & Data: PyTorch, TensorFlow, Scikit-learn, XGBoost, Classification & Pattern Matching, Feature Engineering, Embeddings, Data Preprocessing, Large-Scale Heterogeneous Datasets.

Vector & Semantic Search: Vector Databases (Qdrant, PGVector), Semantic Search, Hybrid Search (SPLADE + Dense), Cross-Encoder Re-ranking, Vector Indexing.

MLOps & Production: Scalable ML Pipelines, Model Monitoring, Low-Latency Inference, API Design (FastAPI, REST), Docker, Kubernetes, Terraform, Temporal.io, CI/CD (GitHub Actions), Kafka.

Cloud & Observability: AWS (SageMaker, Bedrock, Lambda, ECS), Azure OpenAI, GCP; Langfuse, OpenTelemetry, SigNoz, ELK Stack, CloudWatch, Prometheus.

Languages & Security: Python, Java, TypeScript, SQL; Keycloak, Stateless JWT, RBAC (Role-Based Access Control).

PROFESSIONAL EXPERIENCE

Bigstep Technologies

Remote

Senior AI/ML Engineer, Lead Architect

04/2025 – Present

- As **Lead Architect**, set technical direction for cloud-native GenAI delivery across client accounts spanning high-growth startups (Seed to Series F) and global enterprises including **CNBC London** and **VTS**, translating ambiguous problems into production AI architectures in small teams.
- **Agentic Workflows:** Built a **serverless multi-agent** framework with **tool/function calling** over **MCP (Model Context Protocol)** and **swarm-based orchestration** for automated analysis and recommendations. Cut idle compute via a pure **pay-per-execution** model (saving ~\$45k/year).
- **RAG & Semantic Search:** Built a **Hybrid Search** pipeline over technical datasets fusing **SPLADE** (learned sparse) and **Sentence-BERT** (dense) retrieval via **Reciprocal Rank Fusion**, then layered **Cross-Encoder re-ranking**, improving **NDCG@5** by ~15 points (**0.71 to 0.86**).
- **LLM Application (Outcomelabs.ai):** Architected a **Field-Aware Extraction Engine** processing **2M+ records/month** that parses and reasons over unstructured records. A **"Strategy Router"** routes ~60% of fields to deterministic parsers, reserving the LLM for the rest, cutting token spend ~40% with no accuracy loss.
- **Trust & Determinism (VTS):** Built an agentic **document-intelligence** pipeline with **structured-output validation**; diagnosed a **10-point accuracy regression** and redesigned prompt strategy and image preprocessing to recover the full drop, restoring extraction to its ~88% **baseline**.
- **Classification at Scale:** Constructed an entity-matching pipeline on 22M records using an ensemble of **TF-IDF/Cosine Similarity** and 7 text metrics feeding an **XGBoost classifier** (85.3% AUC).

Kuttl (Incubated at IIM Lucknow)

Remote

Founding Engineer

12/2023 – 03/2025

- Joined as **Founding Engineer** for an IIM Lucknow-incubated, GenAI-native orchestration platform; built the MVP from 0-to-1 that secured a term sheet, with a **zero-downtime handover** to the incoming engineering team.
- **Fine-tuned Llama-3-8B** (using LoRA) to replace GPT-4 for structured entity extraction, achieving **98% schema compliance** while reducing per-token inference costs by **90%** and latency by 300ms.
- **Evaluation & Safety:** Operationalized the ML lifecycle with **CI/CD evaluation gates (RAGAS)** scoring **faithfulness and answer-relevancy**, automatically blocking deployments below threshold and catching quality regressions before production.
- **MLOps:** Engineered a distributed document-parsing pipeline using **FastAPI** and **Temporal.io**, processing 50k+ daily messages with resilient retry logic and dead-letter queues to ensure no data loss.

- Applied **semantic caching** and request batching for **RAG** endpoints, improving **P99 latency by 40%** during high-concurrency bursts.

Amplicomm

Bengaluru, India

SDE-2

05/2023 – 12/2023

- Led a team of 3 through a **Strangler-Fig migration** of a legacy monolith to Spring Boot microservices, enabling independent, incremental deploys with instant rollback; enforced **SonarQube** quality gates that cut cyclomatic complexity ~42%.
- Established a centralized identity broker using **Keycloak** and **Spring Security**, replacing stateful DB lookups with **stateless JWT** validation to cut login latency by 30%.
- Engineered a **Schema-per-Tenant** architecture with **RBAC (Role-Based Access Control)** for 21 enterprise clients, ensuring data integrity via automated checksum reconciliation pipelines.

Amazon

Bengaluru, India

Software Development Engineer (SDE)

03/2021 – 03/2023

- Owned the full SDLC for high-scale microservices, co-developing the SmartPOS billing engine and scaling it to support **3,000+ enterprise merchants** with 99.99% availability.
- Optimized cloud infrastructure costs by **25% (~\$100k/yr)** via strategic re-architecture of ECS services to utilize **Spot Instances** and custom auto-scaling policies.
- Developed a data-driven testing framework and **CloudWatch** service-level alarms, identifying **N+1 query bottlenecks** that cut p95 API latency by **20%** and surfaced performance regressions before release.
- Executed the zero-downtime migration of 6 core microservices from GCP to AWS, ensuring data consistency and seamless customer transition.

Publicis Sapient

Remote, India

Associate Data Engineer

09/2020 – 03/2021

- Designed high-frequency data ingestion pipelines using **Apache Airflow** and **Crawlera** to scrape pricing data from Amazon, Walmart, and Target, bypassing complex anti-bot protections for competitive intelligence.
- Refined **Elasticsearch** indexing strategies and aggregation queries, reducing query latency by ~40% to enable sub-second dashboard load times for brand performance analytics.

EDUCATION

National Institute of Technology Delhi

Delhi, India

B.Tech in Computer Science & Engineering

2016 – 2020

ACHIEVEMENTS

Amazon "On the Hero Road" Award: Recognized for exceptional impact on team delivery and engineering excellence.

Udacity Computer Vision Nanodegree: Full scholarship recipient (sponsored by Facebook).

Deep Learning Specialization: Completed 5-course specialization (Coursera) with perfect score.